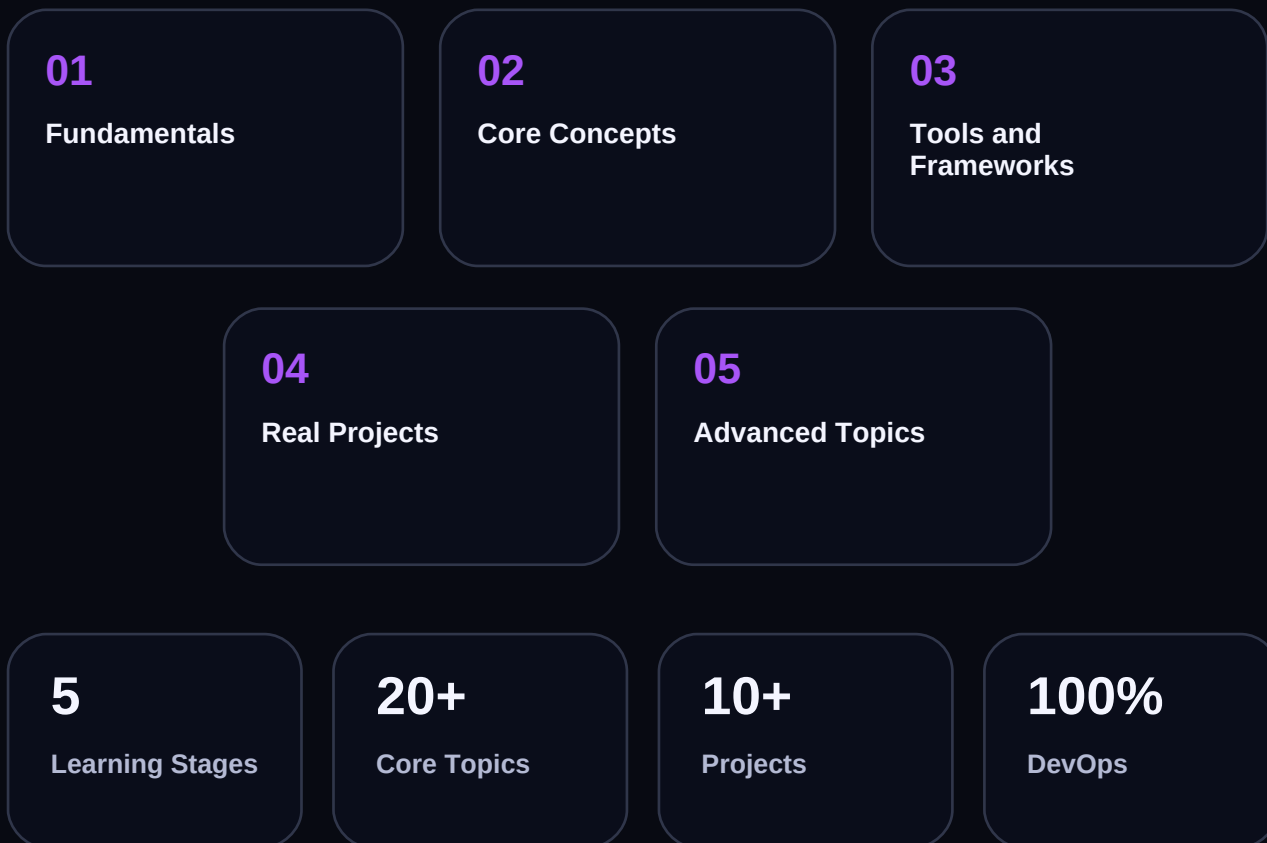
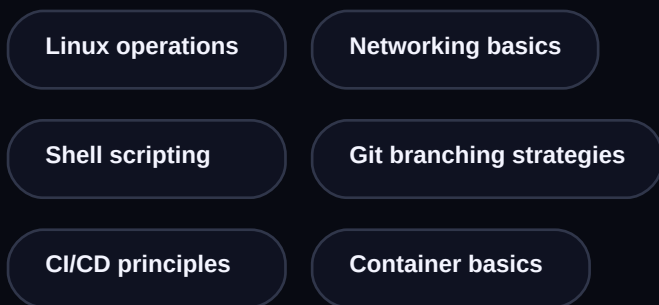




DEVOPS

Complete Learning Roadmap

Automate software delivery, infrastructure reliability, and release confidence at scale.





Learning Path Overview

Five progressive stages from fundamentals to production mastery

01

Fundamentals

Linux operations • Networking basics • Shell scripting • Git branching strategies

02

Core Concepts

CI/CD principles • Container basics • Infrastructure as code concepts • Monitoring foundations

03

Tools and Frameworks

Docker • Kubernetes • Terraform

04

Real Projects

Pipeline for microservices • Blue-green rollout • Incident runbook setup • Autoscaling setup

05

Advanced Topics

SRE metrics • Disaster recovery drills • Security in pipelines • Platform engineering

How to Use This Roadmap

Build and ship first

Complete a small working project in every stage before advancing.

Use official sources

Start with the docs and use secondary resources to reinforce implementation patterns.

Iterate on projects

Refactor earlier work as you learn better architecture, testing, and performance habits.

Stages 01 & 02 - Deep Dive

Fundamentals, core concepts, and architectural foundations

01

Stage 1 - Fundamentals

Estimated: 3-4 weeks | Prerequisites: Basic foundations

Topics

- Linux operations
- Networking basics
- Shell scripting
- Git branching strategies

Tools & Frameworks

- Linux operations
- Networking basics
- Shell scripting
- Git branching strategies

Recommended Projects

- DevOps workshop using Linux operations
- DevOps portfolio project with Networking basics

Stage 1 Milestone Complete DevOps workshop using Linux operations and document the decisions, trade-offs, and validation checks before moving to Stage 2.

02

Stage 2 - Core Concepts

Estimated: 4-5 weeks | Prerequisites: Stage 1 complete

Topics

- CI/CD principles
- Container basics
- Infrastructure as code concepts
- Monitoring foundations

Tools & Frameworks

- CI/CD principles
- Container basics
- Infrastructure as code concepts
- Monitoring foundations

Recommended Projects

- DevOps workshop using CI/CD principles
- DevOps portfolio project with Container basics

Stage 2 Milestone Complete DevOps workshop using CI/CD principles and document the decisions, trade-offs, and validation checks before moving to Stage 3.

Stages 03 & 04 - Deep Dive

Modern tooling, delivery frameworks, and portfolio projects

03

Stage 3 - Tools and Frameworks

Estimated: 4-6 weeks | Prerequisites: Stage 2 complete

Topics

- Docker
- Kubernetes
- Terraform
- Prometheus and Grafana

Tools & Frameworks

- Docker
- Kubernetes
- Terraform

Recommended Projects

- DevOps workshop using Docker
- DevOps portfolio project with Kubernetes

Stage 3 Milestone Complete DevOps workshop using Docker and document the decisions, trade-offs, and validation checks before moving to Stage 4.

04

Stage 4 - Real Projects

Estimated: 6-8 weeks | Prerequisites: Stage 3 complete

Topics

- Pipeline for microservices
- Blue-green rollout
- Incident runbook setup
- Autoscaling setup

Tools & Frameworks

- Pipeline for microservices
- Blue-green rollout
- Incident runbook setup
- Autoscaling setup

Recommended Projects

- DevOps workshop using Pipeline for microservices
- DevOps portfolio project with Blue-green rollout

Stage 4 Milestone Complete DevOps workshop using Pipeline for microservices and document the decisions, trade-offs, and validation checks before moving to Stage 5.



Stage 05 - Advanced Topics & Delivery

Performance, architecture, release automation, and production governance

05

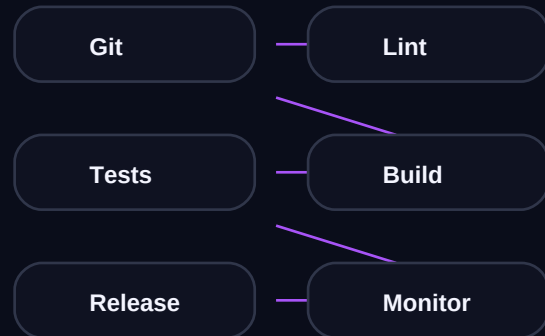
Stage 5 - Advanced Topics

Estimated: Ongoing | Industry-grade operating habits

Advanced Focus

- SRE metrics
- Disaster recovery drills
- Security in pipelines
- Platform engineering

CI/CD Pipeline



Stage 5 Milestone

A production-grade devops workflow with measurable performance, quality, and release confidence.

Performance Budget

Set and track concrete engineering thresholds: startup latency, error budget, throughput, cost efficiency, and deployment confidence.

Resources & Practice Questions

Verified links, self-assessment, and community follow-up

FREE

DevOps docs and community guides

Link: <https://docs.github.com/actions>

FREE

Build one portfolio project per stage before moving ahead

Link: <https://roadmap.sh/devops>

PAID

Structured specialization track focused on DevOps

Link: <https://roadmap.sh/devops>

FREE

DevOps roadmap reference

Link: <https://roadmap.sh/devops>

FREE

Official documentation

Link: <https://docs.github.com/actions>

Self-Assessment Practice Questions

- How would you evaluate readiness for Stage 3 in DevOps?
- Which project best proves Stage 4 competency in DevOps?
- What trade-offs appear in Stage 5 decisions for DevOps?